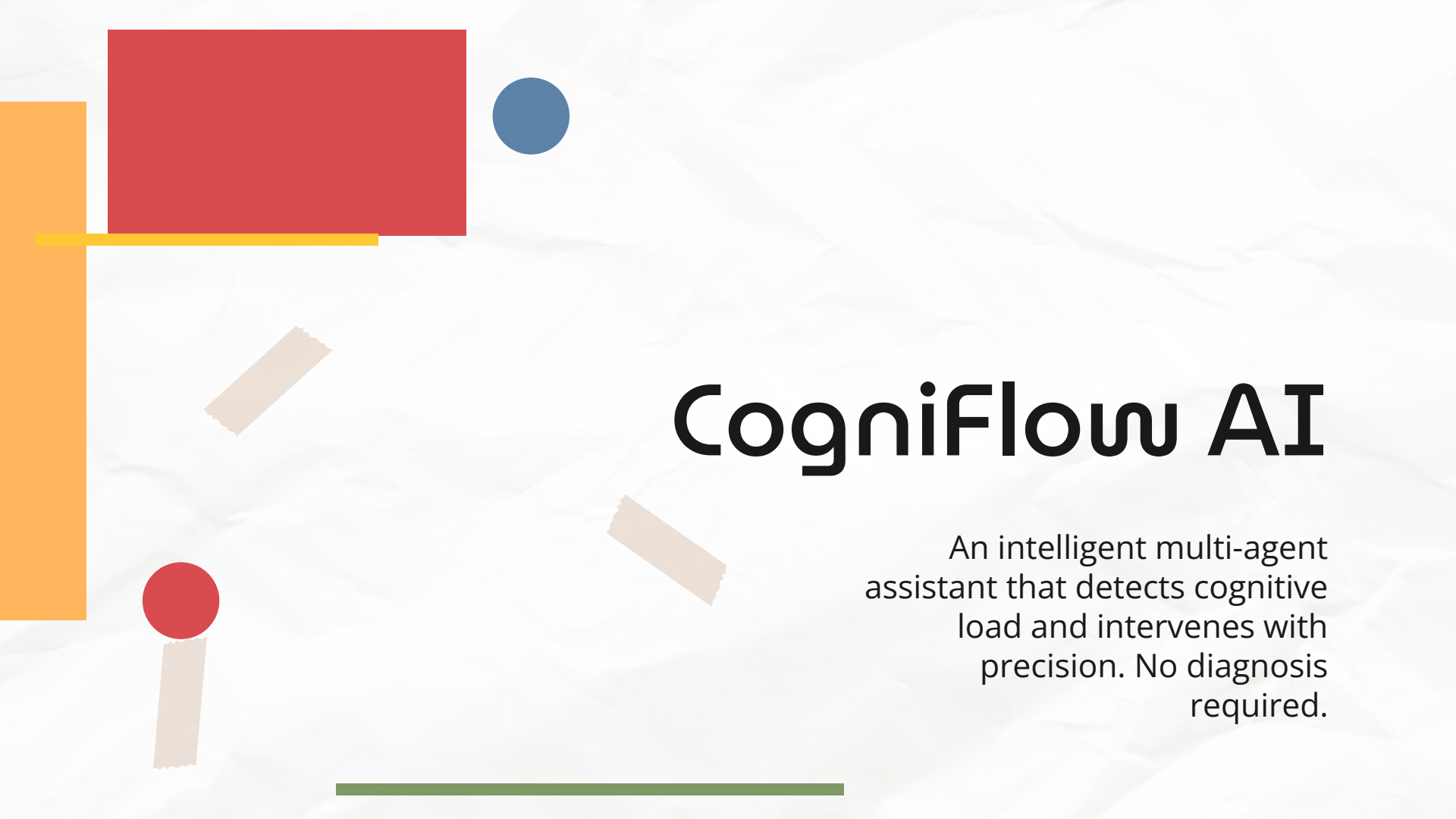


The background is a light gray surface with a crumpled paper texture. On the left side, there is a vertical orange bar. Above it, a red rectangle is partially covered by a yellow horizontal bar. To the right of the red rectangle is a blue circle. Below the red rectangle, there is a red circle with a small piece of torn paper attached to its bottom. Further down and to the right, there are two more pieces of torn paper. At the bottom center, there is a horizontal green bar.

CogniFlow AI

An intelligent multi-agent
assistant that detects cognitive
load and intervenes with
precision. No diagnosis
required.

Problem

Neurodivergent professionals don't struggle with the work itself. They struggle with doing the work while simultaneously performing neurotypicality.

- 1 in 5 tech professionals is neurodivergent
- 30–40% of working memory consumed by masking
- No existing tool detects or addresses this cost

The People We Built This For

Sofia: ADHD · Executive Load

Gets stuck mid-task, loses context, can't re-enter.
"Deficits in behavioral inhibition impair the ability to re-engage with interrupted tasks." Barkley (1997), Executive Functions

MTI proxy: keystroke latency > 2000ms + abandonment signals **MTI baseline: 71**

Ana: Autism · UX Designer

Freezes on ambiguous messages, delays responses for hours.
"Camouflaging autistic traits requires continuous social monitoring, consuming working memory resources otherwise available for task performance." Hull et al. (2017), CAT-Q

MTI proxy: section returns > 4 + high latency variance **MTI baseline: 64**

Miguel: Dyslexia · Data Analyst

Re-reads the same paragraph 4–5 times before acting.
"Phonological loop overload reduces reading comprehension and increases re-reading frequency in adults with dyslexia." Baddeley (2003), Working Memory Model

MTI proxy: high revision frequency + scroll without progression
MTI baseline: 58



CogniFlow Solution

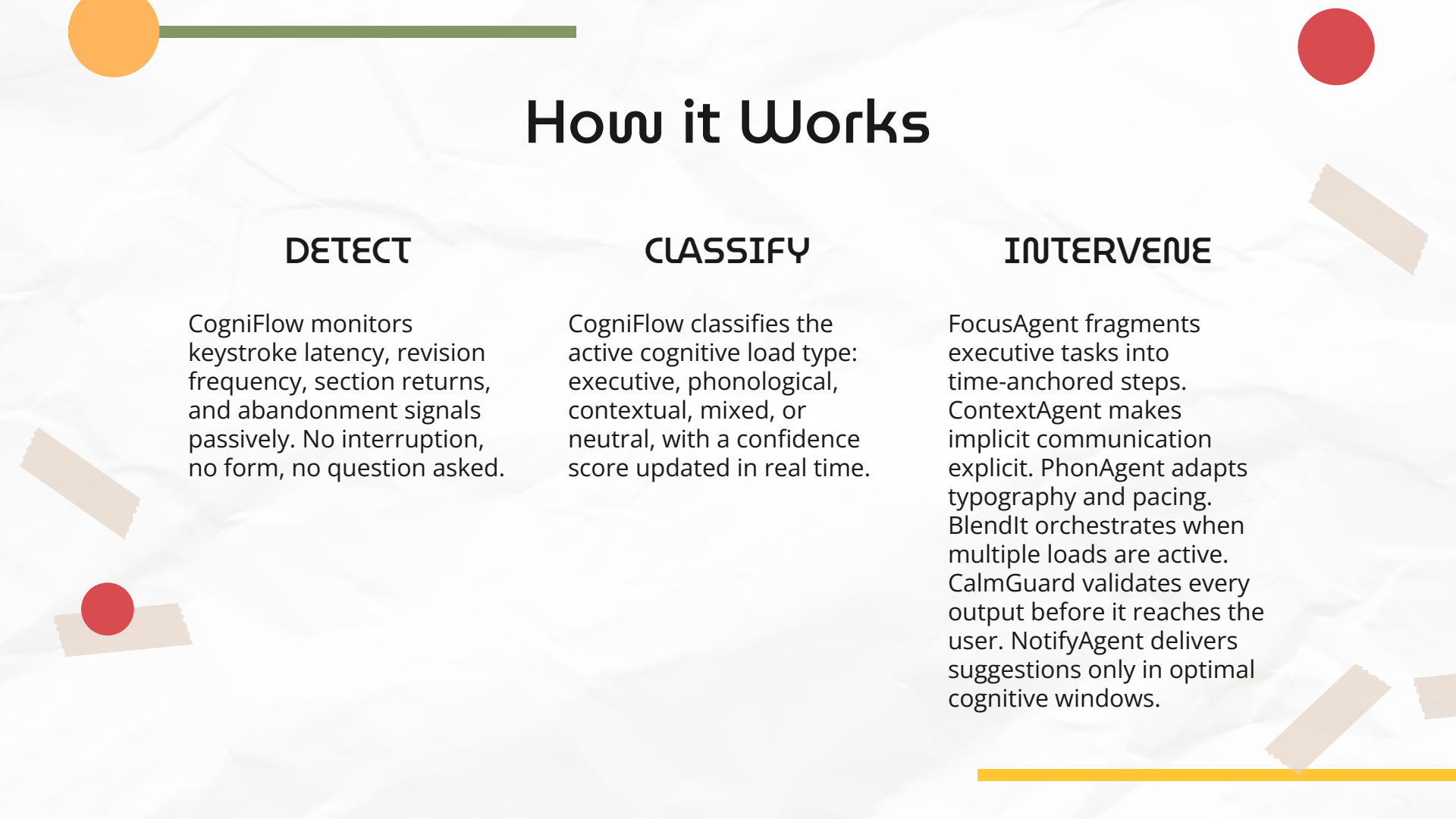
MASKING TAX INDEX

A proprietary metric calculated from behavioral proxies: keystroke latency, revision frequency, section returns, and abandonment signals. No form. No declaration. No diagnosis.

THE APPROACH

7 specialized AI agents monitor, classify, and intervene in real time across three professional surfaces: text editor, inbox, and task manager.





How it Works

DETECT

CogniFlow monitors keystroke latency, revision frequency, section returns, and abandonment signals passively. No interruption, no form, no question asked.

CLASSIFY

CogniFlow classifies the active cognitive load type: executive, phonological, contextual, mixed, or neutral, with a confidence score updated in real time.

INTERVENE

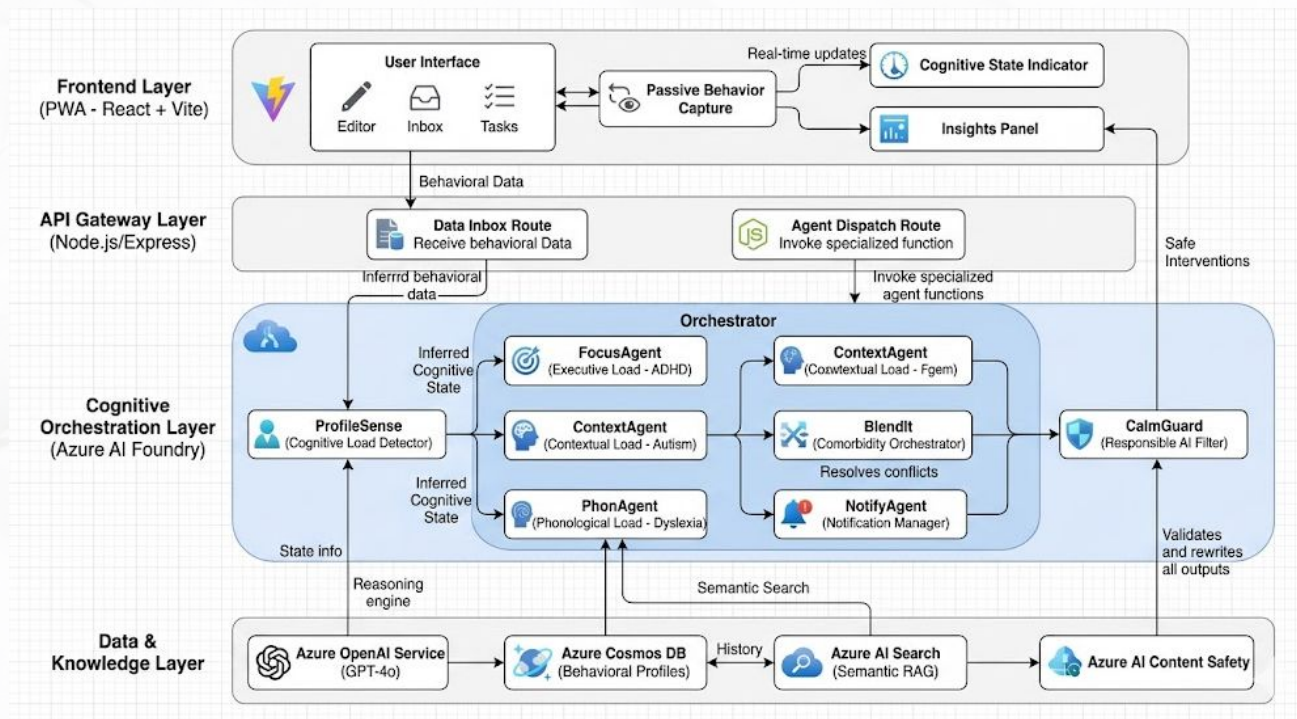
FocusAgent fragments executive tasks into time-anchored steps. ContextAgent makes implicit communication explicit. PhonAgent adapts typography and pacing. BlendIt orchestrates when multiple loads are active. CalmGuard validates every output before it reaches the user. NotifyAgent delivers suggestions only in optimal cognitive windows.



The 7 Agents

- **ProfileSense:** Classifies cognitive load type from behavioral signals, no diagnosis required.
- **FocusAgent:** Fragments executive tasks into time-anchored micro-steps for ADHD profiles.
- **ContextAgent:** Makes implicit communication explicit for autism profiles.
- **PhonAgent:** Adapts typography and reading pacing for dyslexia profiles.
- **BlendIt:** Orchestrates agent conflicts when multiple cognitive loads are active simultaneously.
- **CalmGuard:** Validates every agent output before it reaches the user. RAI structural layer.
- **NotifyAgent:** Delivers interventions only in optimal cognitive windows. Maximum 4 per day. Blocked when MTI > 60.

Architecture

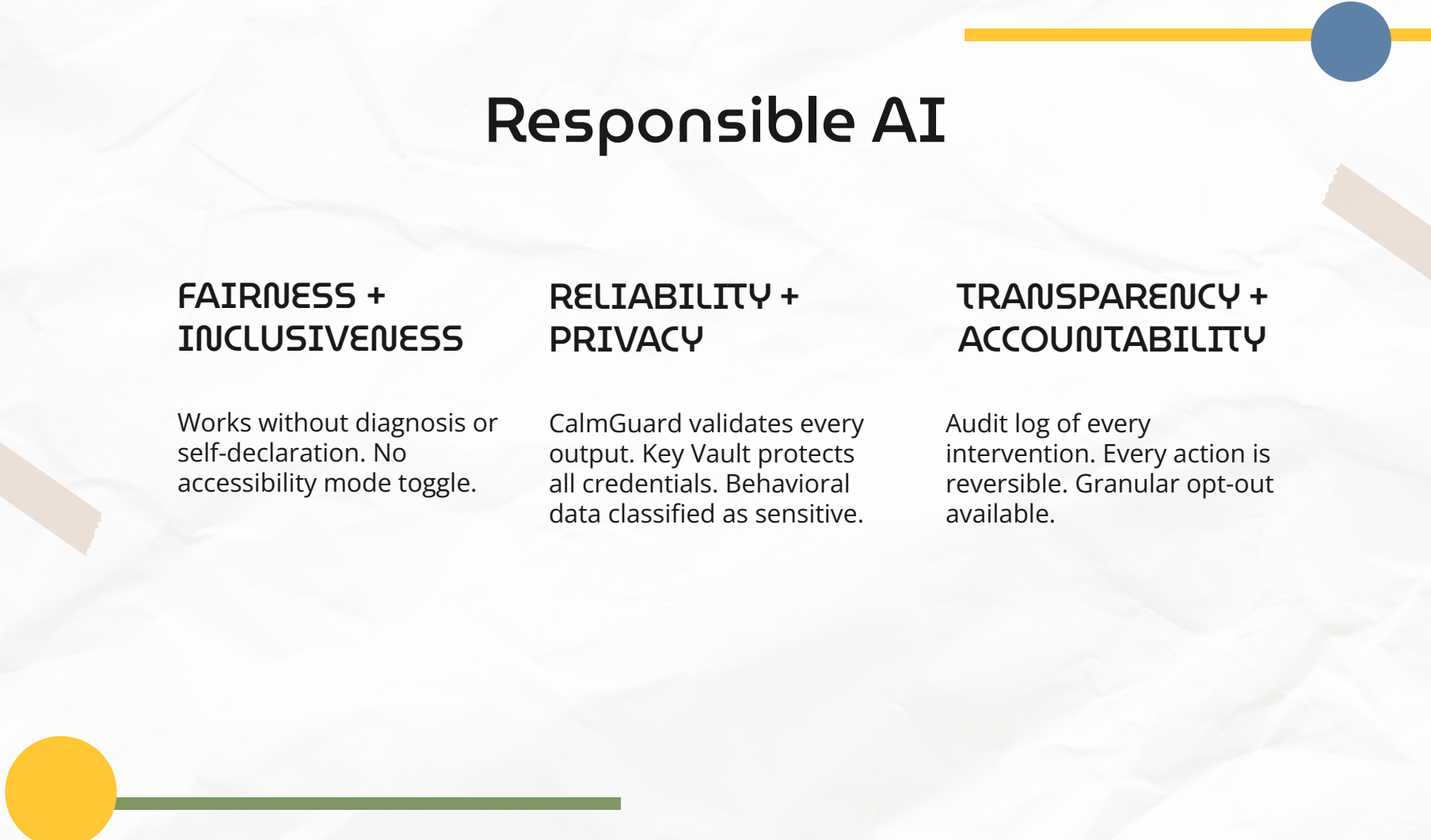




Azure Services



- Azure AI Foundry · Central hub for all 7 agents
- Azure OpenAI · GPT-4.1 mini inference engine
- Azure AI Search · RAG + behavioral index enrichment
- Azure Cosmos DB · Behavioral profiles + session events
- Azure Content Safety · CalmGuard real-time filter
- Azure Notification Hubs · Web Push delivery via NotifyAgent
- Azure API Management · Gateway + rate limiting
- Azure Key Vault · Credentials and secrets management
- Azure Monitor · RAI audit log + intervention tracking
- Azure App Service · Backend + PWA deployment



Responsible AI

FAIRNESS + INCLUSIVENESS

Works without diagnosis or self-declaration. No accessibility mode toggle.

RELIABILITY + PRIVACY

CalmGuard validates every output. Key Vault protects all credentials. Behavioral data classified as sensitive.

TRANSPARENCY + ACCOUNTABILITY

Audit log of every intervention. Every action is reversible. Granular opt-out available.

Impact

**-52% MTI
reduction
per session
(Sofia · editor)**

**-50% Response
decision time
(Ana · inbox)**

**0 Diagnoses
required**

CogniFlow gives back the working memory that masking steals.



Thank You!

CogniFlow AI

Erika Barrado
Letícia Batista Silva